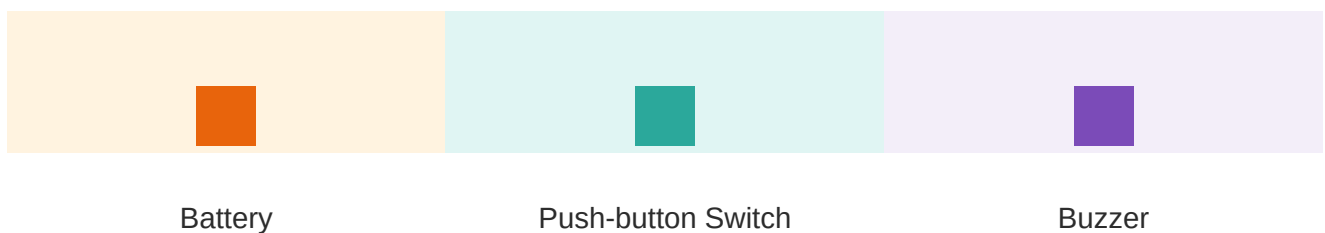


■ Grade 3 · Hardware Lesson

Push-button Switch & Buzzer Circuit

Learn how a push-button and buzzer work together in a simple series circuit!



What Will We Learn Today?

1. What is a circuit?
2. Open vs Closed circuits
3. Meet the push-button switch
4. Meet the buzzer
5. Parts we need
6. Circuit diagram
7. Step-by-step: build it!
8. What happens when you press?
9. Real-life examples
10. Safety rules

1. What is a Circuit?

A **circuit** is a path that electricity flows through. For electricity to flow, the path must be **complete** — there can be no gaps.

■ Power Source	Provides the electrical energy (e.g. a battery).
■ Wires	Carry electricity from one component to another.
■ Component	Uses the electricity to do something useful (light, sound...).

Key idea: When the path is complete, electricity flows!

2. Open vs Closed Circuit

■ OPEN Circuit There is a gap in the path. Electricity cannot flow. Component is OFF.	■ CLOSED Circuit The path is complete. Electricity flows freely. Component is ON!
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■ A **switch** controls whether a circuit is open or closed.

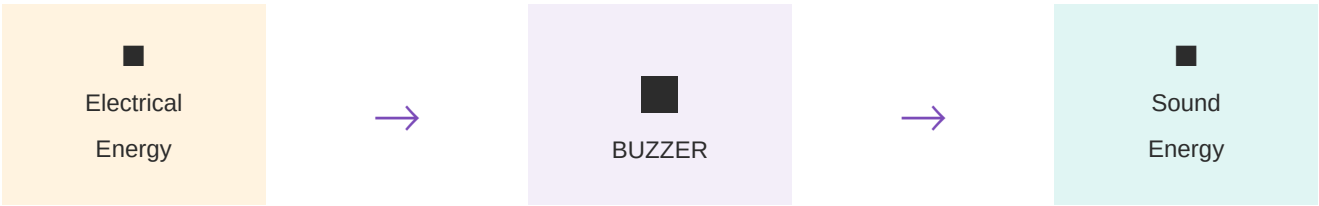
3. Meet the Push-button Switch!

A **push-button switch** is a *momentary* switch — it only closes the circuit **while you press it**. The instant you let go, the circuit opens again.

Inside	Two metal contacts .
Press	Contacts touch → circuit closed → current flows!
Release	Contacts separate → circuit opens → no current.

4. Meet the Buzzer!

A **buzzer** converts **electrical energy** into **sound energy**. When electricity flows through it, it vibrates and produces a beeping sound.



The buzzer has a + **(positive)** pin and a – **(negative)** pin. Connect them the right way around!

5. Parts We Need

1	■ Battery	3 V or 9 V
2	■ Push-button Switch	Momentary type
3	■ Buzzer	Active buzzer
4	■ Connecting Wires	Red (+) and Black (–)
5	■ Breadboard	Optional — makes wiring easier

6. Circuit Diagram

The components are connected in a **series circuit** — electricity flows through each one in turn.



Battery (+) → Wire → Push-button → Wire → Buzzer → Wire → Battery (-)

7. Let's Build It! — Step by Step

Step 1

Connect the **RED wire** from the **battery (+)** terminal to one leg of the **push-button switch**.

■ *Red = Positive (+)*

Step 2

Connect another wire from the **other leg** of the push-button to the **(+) pin** of the buzzer. The switch is now sitting in the middle of the circuit!

■ *Switch is in the middle*

Step 3

Connect the **BLACK wire** from the **buzzer (-)** pin back to the **battery (-)** terminal. The circuit is now complete!

■ *Black = Negative (-)*

8. What Happens?

■ Press the button

Circuit CLOSSES → electricity flows → buzzer goes BEEP! ■

■ Release the button

Circuit OPENS → no electricity → buzzer goes silent. ■

You control the sound with your finger! ■

9. Real-life Examples

■ Doorbell

Press → ring! A classic push-button + buzzer circuit.

■	Game Show Buzzer	Players press to buzz in with an answer.
■	Elevator Buttons	Each floor button is a push-button switch.
■ ■	Keyboard Keys	Every key you type uses a tiny push-button switch.
■	Alarm Systems	Buttons trigger alerts or emergency signals.

10. Safety Rules ■ ■

- Use only low-voltage batteries (3 V – 9 V) for this experiment.
- NEVER connect your circuit to a wall socket or mains electricity.
- Ask an adult if you are unsure about any connection.
- Handle wires carefully — avoid sharp or frayed ends.

Great Job! Today You Learned...

■	Push-button	A momentary switch — closes only while pressed.
■	Buzzer	Converts electrical energy into sound energy.
■	Series Circuit	All components are in one loop; each is needed.
■	Open/Closed	A switch controls whether electricity can flow.

■ **Challenge:** Can you add an LED to the circuit so it lights up at the same time as the buzzer beeps?